

Total no. of Pages....

Roll no.

II SEMESTER B.Tech.

END TERM EXAMINATION, April-May.-2026

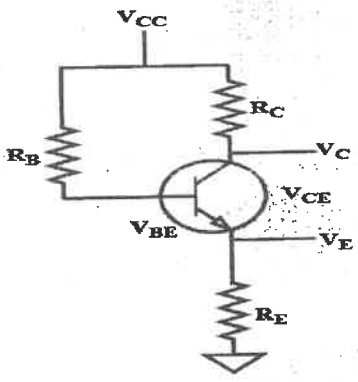
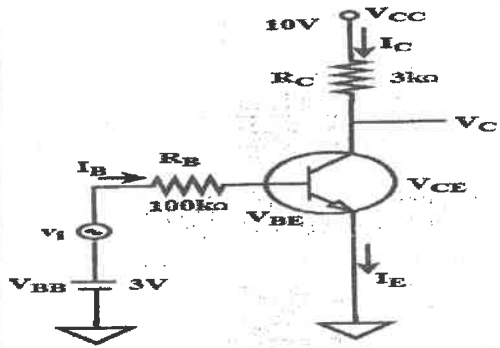
Paper Code – EC/VT/IC ECC202

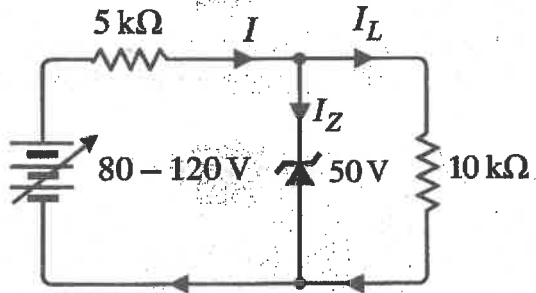
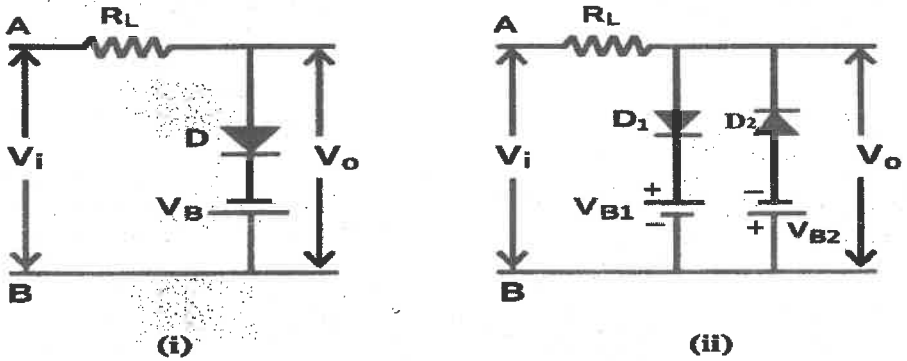
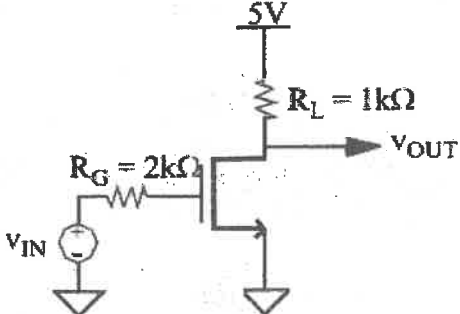
Subject: Electronic Devices & Circuits

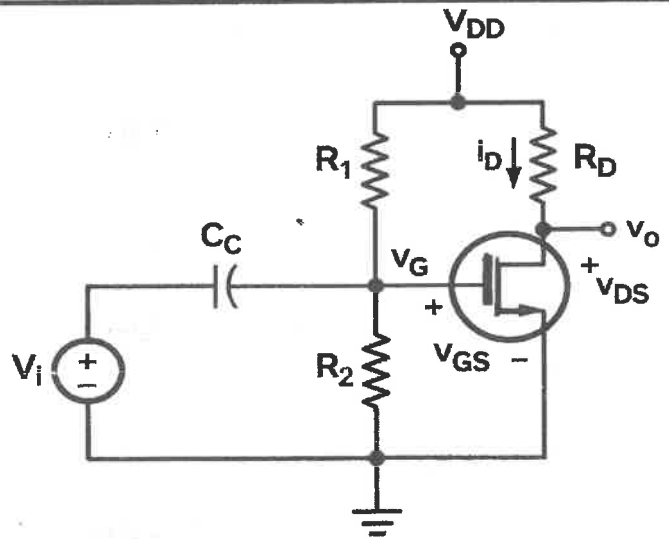
Time: 3 Hours

Max. Marks - 50

Note: - Missing data/ information if any, may be suitably assumed & mentioned in the answer. All questions carry equal marks [10=5+5].

	There are three parts in each question. Attempt any two parts.	CO
1a	For following circuit given that: $V_C = 7.6\text{ V}$, $V_{CC} = 12\text{ V}$, $I_C = 3\text{ mA}$, $V_E = 2.4\text{ V}$, If $\beta = 80$, Determine and calculate (a) R_C (b) R_E (c) R_B (d) V_{CE} (e) V_B 	CO ₁
1b	Using small signal model, calculate voltage gain, R_{in} and R_{out}. ($\beta=100$) 	CO ₁
1c	Draw and briefly explain any two circuits with proper labelling: - Current mirror - Unity voltage gain circuit - Ebers-Moll model	

2a	A 100 V, 50 Hz AC supply is connected to a bridge rectifier with a load resistance of $2\text{ k}\Omega$ and transformer turn ratio 10:1, Calculate DC Load Current (I_{dc}), rms current (I_{rms}), ripple factor and efficiency of the bridge rectifier. Assume ideal diodes.	CO ₁
2b	Find the maximum and minimum values of Zener diode current. 	CO ₂
2c	Determine output voltage and draw output waveform for given circuits if $V_i = 5\sin(100t)$ and $V_B = 2\text{V}$, assume ideal diodes: 	CO ₂
3a	Derive the expression for the drain current in a MOSFET. Also, sketch the V-I characteristics and clearly indicate the different regions of operation.	CO ₄
3b	Compute small signal parameters g_m and dc voltage V_{GS} and V_{out} of this amplifier with $\mu_n C_{ox} = 100\text{ }\mu\text{A/V}^2$, $(W/L) = 6/0.1$, $V_t = 0.5\text{ V}$, $I_D = 0.275\text{ mA}$. 	CO ₃
3c	Derive expression for voltage gain (A_v), input resistance (R_{in}) and output resistance (R_{out}) for the given circuit.	

		
4a	A semiconductor has electron concentration $n = 5 \times 10^{15} \text{ cm}^{-3}$. The drift velocity of electrons is $3 \times 10^5 \text{ cm/s}$ under an electric field of 300 V/cm. (i) Calculate electron mobility (ii) Determine the drift current density.	
4b	Calculate the barrier potential of a step-graded PN junction. Also, state the Shockley diode equation and explain the significance of each term involved.	CO ₃
4c	Explain the concepts of drift current and diffusion current in semiconductors. Also, sketch and describe the V-I characteristics of an ideal diode, silicon diode, LED, and Zener diode.	CO ₄
5a	Explain why an operational amplifier is so named. State its ideal characteristics and discuss the concepts of virtual ground and CMRR.	
5b	Derive the expressions for the voltage gain of inverting and non-inverting amplifiers with help of neatly drawn circuits. Determine suitable resistor values to obtain a voltage gain of 100 in each case.	
5c	(i) The input voltage to the circuit is given by $v_i = 0.1 \sin 2000\pi t \text{ V}$. What is the output voltage if $R = 10 \text{ k}\Omega$, $C = 0.005 \text{ }\mu\text{F}$, and $v_o(t=0) = 0$? (ii) What is the voltage transfer $V_o(s)/V_i(s)$ function? 